PREET MISTRY

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EDUCATION

University of Toronto **September 2018 - May 2024**
Bachelor's, Computer Engineering *GPA: 3.21*

PROFESSIONAL EXPERIENCE

Career Break: Travel, Volunteering & Independent Projects **October 2025 - September 2026**
Independent Software Developer & Community Volunteer

- Engineered and deployed a full-stack, mobile, workout tracking app (see Projects)
- Volunteered 3-4 months on a Buddhist monastery's snow and grounds safety team, proactively coordinating snow and ice response (snow blowers, clearing trucks, shoveling, salting) to keep pathways safe for residents and visitors
- Traveled internationally and supported family commitments, including a family wedding

Amazon **May 2025 - September 2025**
Software Development Engineer *Toronto, ON, Canada*

- Implemented validation guardrails for configuration management in Alexa+ LLM API infrastructure using AWS Lambda, emphasizing scalability and reducing adverse events during dynamic inference by 3x
- Diagnosed and fixed production bugs in asynchronous agentic LLM system by tracing path through execution graph in spite of limited logging availability, causing a drop in bug reports by 10%
- Contributed to architecture migration plan of agentic LLM monolith into microservices, proposing broader and deeper inference branching to enable parallel action exploration while reducing latency

MSRG - University of Toronto **May 2023 - February 2025**
Deep Learning Researcher *Toronto, ON, Canada*

- Pioneered a Graph Neural Network (GNN) architecture for molecular bond strength prediction, rivaling existing literature benchmarks with an MAE score of 0.50
- Meticulously optimized training and hyperparameter tuning routines, slashing compute resource utilization by >30x while maintaining model performance
- Architected a robust execution methodology that reduced 95% of runtime failures in distributed computing environments

Ultimate Kronos Group (UKG) **August 2021 - August 2022**
Full Stack Developer *Toronto, ON, Canada*

- Successfully delivered a highly secure web application for client management and provisions, enhancing administration efficiency and mitigating vulnerabilities
- Conceptualized a responsive frontend that minimized user error during critical service changes using TypeScript/Angular
- Collaborated with a cross-functional team to design and rigorously test a Java-based containerized backend, reducing potential security risks
- Integrated the web application into CI/CD environment, utilizing Docker and Kubernetes to ensure streamlined deployment

PROJECTS & OUTSIDE EXPERIENCE

Fitness Tracking Mobile Full Stack App **March 2026 - October 2026**

- Designed and developed a multi-component, polyglot system (Rust, TypeScript, SQL) spanning a Tauri-based client with an async Rust core and an Axum/Postgres REST backend deployed via containerization and verified across Android and desktop
- Engineered a resilient synchronization engine in async Rust: guarded against overlapping cycles and sign-out races, kept state transitions atomic under network failure, and minimized bandwidth utilization with checksum-based reconciliation
- Secured REST API with JWT auth (custom extractor), argon2 hashing, timing-attack and user-enumeration mitigations, and token-derived, per-user query scoping

SKILLS

Skills: Java, TypeScript, JavaScript, HTML/CSS, Python, SQL, Postgres, MySQL, React.js, Angular, Testing, Agile, REST APIs, C/C++, Rust, AWS, Docker, Kubernetes, Async, Security and Identity, DevOps, Git, Google Cloud Platform, Next.js, Linux/Unix